Water Jug Problem

Water Jug Problem.py - C:/Users/Vineela/AppData/Local/Programs/Python/Python315/Water Jug Problem.py (...

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```
from collections import deque

start = (0, 0)
goal = 2
queue = deque([(start, [start])])
visited = {start}

while queue:
    (a, b), path = queue.popleft()

    if a == goal:
        print("Water Jug Solution:")
        for state in path:
            print(state)
        break

    pour1 = min(a, 3 - b)
    pour2 = min(b, 4 - a)

    next_states = [
        (4, b),
        (a, 3),
        (0, b),
        (a, 0),
        (a - pour1, b + pour1),
        (a + pour2, b - pour2)
    ]

    for state in next_states:
        if state not in visited:
            visited.add(state)
            queue.append((state, path + [state]))
else:
    print("No solution found.")
```

IDLE Shell 3.15.0

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```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Vineela/AppData/Local/Programs/Python/Python315/Water Jug Pr
oblem.py
Water Jug Solution:
(0, 0)
(4, 0)
(1, 3)
(1, 0)
(0, 1)
(4, 1)
(2, 3)
>>>
```